

2026-06-10-impl-batch5-review

Wave-5 Code-Review Report (§11.4.125 / §11.4.134)

Reviewer: Independent code-review subagent (Opus 4.8) **Date:** 2026-06-10 **Scope:** UNCOMMITTED wave-5 batch — SP4-cont (submodules/helix_agent/internal/clis) + SP7 (helix_code/tests/{ddos,scaling,ux,ui} + scripts/{gates,tests}/*) **Mandate:** read-only verification; no edits/commits/push. Mutations applied during review were reverted (restore confirmed byte-identical, working tree clean).

VERDICT: GO

Zero BLOCKING findings. 2 NON-BLOCKING notes (advisory). All required builds/tests/gates/meta-tests run GREEN; every §1.1 mutation FAILED the guard as required; restores confirmed clean.

- BLOCKING (must-fix): 0
 - NON-BLOCKING (advisory): 2
-

1. Builds + vet (PASS)

```
helix_agent: go build ./internal/clis/... -> EXIT=0 (PASS)
helix_agent: go vet   ./internal/clis/... -> EXIT=0 (PASS)
helix_code:  go vet   ./tests/ddos/... ./tests/scaling/... ./tests/ux/... ./
helix_code:  go vet (re-run, full SP7)      -> VET_OK (PASS)
```

2. SP4-cont RED→GREEN + §1.1 mutations (PASS)

```
helix_agent: go test ./internal/clis/... -count=1 -> ok (all pa
GREEN guards (default):
  TestD12_RealExec_ExecsRealBinary          -> PASS (qwencoder/github
  TestD12_RealExec_AbsentBinaryIsHonestError -> PASS (absent binary =>
  TestD12_HonestError_NoFakeSuccess (40 agents) -> PASS (each returns hon
  TestD11_IsAgentTypeAvailable_RealLookPath  -> PASS (real exec.LookPa
RED reproductions (default => SKIP; PIN_STUB_BLUFF=1 => RUN):
  TestD12_StubsAreBluffs          default -> SKIP (SKIP-OK: ATM-SP4-D12); P
  TestD11_AllowAllTypesBluff      default -> SKIP (SKIP-OK: ATM-SP4-D11); P
§1.1 mutation A (executeQwenCoder real-exec -> "Qwen Coder execution compl
-> TestD12_RealExec_ExecsRealBinary/qwencoder FAIL ("did NOT exec the ag
-> restore -> ok ✓
§1.1 mutation B (executeKiro honest-error -> "Kiro execution completed" fa
-> TestD12_HonestError_NoFakeSuccess/kiro FAIL ("returned success with n
-> restore -> ok ✓
Bluff-literal scan (non-test source):
```

```
grep 'execution completed' internal/clis/*.go (non-test) -> only //-comm
grep 'simulated|for now|TODO implement|in production this would' (non-te
```

The D-11 split is correct and honest: `agentDefaultCommand` maps 8 agents to confirmed CLI binaries (`aider/claude/codex/cline/openhands` + `qwen/copilot/gemini`) and 40 to empty-command honest-error class; `IsAgentTypeAvailable` is a real `exec.LookPath` honoring `HELIX_AGENT_BIN_<TYPE>`. The 3 real-exec D-12 methods go through `runCLIAgent` → real `exec.CommandContext` with real exit-code + `CombinedOutput` (BLUFF-003 satisfied).

3. §11.4.120 reconciled tests — NOT weakened to tautologies (PASS)

`instance_manager_test.go` `TestInstanceManager_IsAgentTypeAvailable` was reconciled, not faked: - old `assert.True(...TypeKiro/TypeContinue)` rewritten to `assert.False(...)` matching the corrected honest contract (those agents genuinely have no resolvable binary); - real-CLI types proven available by injecting a real fake binary via `HELIX_AGENT_BIN_<TYPE>` (not a hardcoded true); - `CreateInstance` + benchmark reconciled to inject a fake aider binary since `CreateInstance` now gates on the real availability check.

```
TestInstanceManager_IsAgentTypeAvailable -> PASS
TestInstanceManager_CreateInstance       -> PASS
```

4. SP7 harnesses + meta-tests (PASS)

```
helix_code: go test ./tests/ddos/... ./tests/scaling/... ./tests/ux/... ./
ok dev.helix.code/tests/ddos          7.619s
ok dev.helix.code/tests/scaling       6.205s
ok dev.helix.code/tests/ux            18.139s
ok dev.helix.code/tests/ui             0.998s
```

Go meta-tests (each plants a defect, asserts harness DETECTS it):

```
ddos:    Detects5xxStorm / DetectsLatencyBomb / DetectsNoServedResponses
scaling: DetectsFlatThroughput / DetectsDegradation / RejectsBelowFloor
ux:      DetectsCannedString / DetectsLeakedID / DetectsEmptyAndTerse /
ui:      DetectsBlankRender / DetectsDeadKeyHandler / DetectsLeakedID /
```

Bash gates+meta-tests CONST-068 (bash -n on all 8): ALL OK (shebang #!/usr

Bash meta-tests (gate FAILs bluff, PASSes real evidence):

```
CM-DDOS-HITS-HELIXCODE  META-TEST PASS (EXIT=0)
CM-SCALING-HITS-HELIXCODE META-TEST PASS (EXIT=0)
CM-UX-HITS-HELIXCODE    META-TEST PASS (EXIT=0)
CM-UI-HITS-HELIXCODE    META-TEST PASS (EXIT=0)
```

Verified the specific anti-bluff claims: - **ux really shells the cli binary** —

`ux_journey_test.go`: `TestUX_CLI_Journey_RealBinary` builds the real `./cmd/cli` then `exec.CommandContext`s it, asserting a per-run nonce echoes through real stdout (BLUFF-003 path); honest SKIP only when go toolchain absent or build fails. - **scaling shows real N-scaling numbers (not hardcoded)** — `scaling_harness.go` drives the REAL `internal/worker.WorkerPool` across `N={1,2,4,8}` via real `RegisterWorker/AssignTask/ReleaseWorker/GetPoolStats`; `GainAtMaxN` is computed from MEASURED throughput-per-second, FAILing if gain < 1.5x (flat) or throughput degrades. Honest boundary documented (in-process concurrent-capacity scale-out; SSH horizontal scale-out is a separate integration-tagged path that SKIPS when no SSH workers configured). - **ddos**

429 assertion correctly gated OFF — `ddos_harness.go:18-19,288-294`:

`internal/server` wires no rate-limit middleware today, so the 429-refusal assertion ships behind `DDOS_EXPECT_RATELIMIT=1` (OFF by default). Default assertions are graceful-degradation (zero 5xx, real served body-marker hits > 0, bounded p99). It never asserts a 429 the code cannot produce. The integration driver (`//go:build integration`) boots the `REAL server`. New with real PG/Redis and honest SKIP when infra unreachable. - **ui** — `ui_harness.go` renders `REAL` `tview/tcell` components into a `tcell.SimulationScreen` cell grid and reads back exact composited cells (genuine pixel-equivalent evidence); Fyne native-pixel layer honestly tracked operator-attended.

5. Anti-bluff + scope (PASS)

- Honest errors genuinely returned (not swallowed): `noHeadlessCLIError` returns a real error; `runCLIAGENT` surfaces non-zero exit / spawn failure as a real wrapped error with the binary's actual output. ✓
- No `NEW simulated|for now|TODO` implement in production source — only pre-existing-style `//comment` citations describing the removed bluff (constitution smoke excludes these). ✓
- §11.4.120 reconciled tests not weakened to tautologies (see §3). ✓
- No `go.mod/go.sum` edits in either repo. ✓
- SP4-cont touches only `helix_agent/internal/clis` (3 files). ✓
- SP7 touches only `helix_code/tests/{ddos,scaling,ux,ui}/ + scripts/{gates,tests}/`. ✓ `qa-results/` is gitignored. ✓
- No secrets logged — only test fixtures: `TEST_PG_PASSWORD` default “helix” (local stack), empty Redis pw, JWT literal explicitly “`ddos-flood-test-secret-not-a-real-credential`”. These are unit/integration test creds, permitted. ✓
- Working-tree quiescence (§11.4.84): after mutation experiments, restore is byte-identical; mutation-residue scan finds NO `MUTATION/always pass` markers. ✓

NON-BLOCKING advisory notes

N1 (advisory) — **RED reproduction polarity is “skip-on-fixed / pass-on-pinned”, not literal RED-FAIL-on-broken-artifact.** `TestD12_StubsAreBluffs / TestD11_AllowAllTypesBluff (PIN_STUB_BLUFF=1)` PASS on the fixed artifact via `early-return-on-honest-error(if err != nil { return // bluff gone })`. This is a valid §11.4.115 polarity-switch pattern — they would FAIL only if a stub re-emitted the literal — and the standing GREEN guards (`TestD12_RealExec_*`, `TestD12_HonestError_*`, `TestD11_IsAgentTypeAvailable_RealLookPath`) plus the confirmed §1.1 mutation FAILs are the real bug-catchers. No action required; noted only because the prompt framed the RED tests as “SKIP on fixed artifact” — under PIN they RUN and pass rather than skip, which is the correct §11.4.115 semantics.

N2 (advisory) — `TestD12_StubsAreBluffs/realexec/github_copilot` **took ~25s under `PIN_STUB_BLUFF=1`** because the host has a real `copilot` binary on `PATH` that `runCLIAGENT` actually exec'd (the test injects no fake binary in the RED path, so the real CLI ran to its own timeout). Harmless (the reproduction still passes) but slow; if the suite runs in CI on a host with a real `copilot/qwen/gemini`, consider bounding via the instance `Config.Timeout` or injecting a fast fake in the RED path. Not a correctness defect.

Captured build/test PASS-FAIL summary

helix_agent	go build ./internal/clis/...	PASS
helix_agent	go vet ./internal/clis/...	PASS
helix_agent	go test ./internal/clis/... -count=1	PASS
helix_agent	D11/D12 GREEN guards (default)	PASS
helix_agent	RED reproductions (PIN_STUB_BLUFF=1)	PASS (ru
helix_agent	§1.1 mutation A (qwen real-exec->literal)	FAIL-as-
helix_agent	§1.1 mutation B (kiro honest-error->fake)	FAIL-as-
helix_agent	reconciled instance_manager_test.go	PASS
helix_code	go vet ./tests/{ddos,scaling,ux,ui}/...	PASS
helix_code	go test ./tests/{ddos,scaling,ux,ui}/... -count=1	PASS
helix_code	go meta-tests (all 4 packages)	PASS (ea
scripts	bash -n (8 gate+meta scripts, CONST-068)	PASS
scripts	4 bash meta-tests (CM-{DDOS,SCALING,UX,UI})	PASS (ga
scope	no go.mod/go.sum edits; paths confined; no secrets	PASS

FINAL VERDICT: GO — 0 blocking, 2 advisory. Batch is anti-bluff-sound and ready to proceed to pre-build sweep + main build.